Design and Physical Implementation of a Noble Schmitt Trigger-Based 12T SRAM Cell in 90-nm Technology

Ankitha A. J., Dileep Kumar A., Sinchana L., Kusha H. P
Department of Electronics and Communication Engineering, Maharaja Institute of Technology, Mysuru, Karnataka, India
Guide: Mrs. Shwetha K

Abstract— This paper presents the design study and physical implementation of a Noble Schmitt Trigger-based 12-transistor (12T) static random-access memory (SRAM) bit cell in a 90-nm GPDK technology environment. The cell combines a cross-coupled storage latch with additional feedback and access devices intended to improve data stability, particularly during read operation. Hold, read and write behavior is examined at transistor level, and static noise margin (SNM) is evaluated using DC butterfly curves. The reported margins are 309.08 mV (RSNM), 430.81 mV (HSNM) and 664.60 mV (WSNM). The memory organization is 512 words × 8 bits (4 Kbit). The digital memory flow includes functional waveform inspection, RTL synthesis, floorplan/core setup, placement, power-distribution planning, signal routing, timing review and generation/viewing of a GDS layout. The supplied implementation summary reports 906 standard cells, area 413867.378 (unit not stated), total power 100.66 mW and leakage power 1.637 µW. The captured hold analysis reports zero violating paths. Bit-cell simulation values are distinguished from array-level implementation metrics, and no unverified DRC/LVS or complete sign-off claim is made.

Index Terms— SRAM, Noble Schmitt Trigger, 12T bit cell, 4-Kbit memory, 90-nm technology, static noise margin, Cadence Virtuoso, Genus, Innovus, physical design.

I. INTRODUCTION

Static random-access memory (SRAM) is widely used in processors, controllers and embedded systems because it provides fast access without periodic refresh. A conventional six-transistor (6T) SRAM bit cell consists of two cross-coupled inverters and two access transistors. Its compact structure is attractive for density, but the shared read/write access path can disturb an internal storage node during a read. This creates a trade-off among read stability, writeability, area and power [1], [2].

As transistor dimensions are scaled, device variation and reduced operating voltage can further reduce memory-cell stability. Static noise margin (SNM) provides a voltage-based measure of the disturbance required to flip the stored state. The butterfly-curve method provides a graphical approach to evaluate this margin [1]. Studies of intrinsic device fluctuations explain why stability must be considered carefully in scaled SRAM designs [2].

A Schmitt Trigger exhibits hysteresis: its switching threshold depends on the direction of the input transition. Positive feedback can reject small input disturbances and improve noise immunity. Schmitt-trigger-based SRAM topologies use this behavior to strengthen the storage response, although additional devices increase cell complexity and can introduce trade-offs in write operation and area [6], [7].

This work documents a Noble Schmitt Trigger-based 12T cell and a 512 × 8 memory implementation. It consolidates the supplied transistor-level stability results and digital physical-design evidence in a 90-nm technology flow. Reported values are retained as supplied; unreported performance is not inferred.

II. CELL ARCHITECTURE AND MEMORY ORGANIZATION

A. 12T Bit-Cell Structure

The proposed bit cell contains twelve transistors. Four devices (P1, N1, P2 and N2) form the cross-coupled storage latch. Four NMOS access transistors (N5–N8) provide access paths, while P3, P4, N3 and N4 form the additional stability-enhancement network. The feedback devices are intended to reinforce the internal state and reduce susceptibility to read disturbance. The schematic was created in Cadence Virtuoso using the 90-nm GPDK device environment.

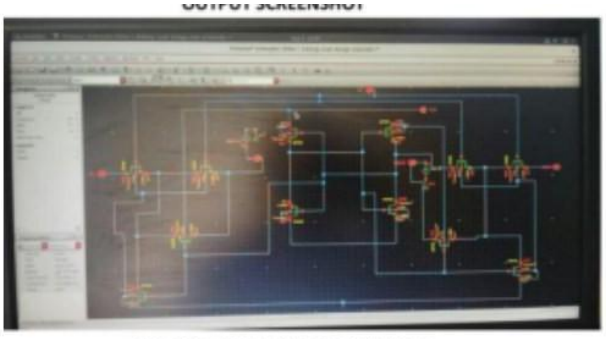


Fig. 1. Transistor-level schematic of the proposed Noble Schmitt Trigger-based 12T SRAM bit cell.

B. Memory Organization and Operating Modes

The memory is organized as 512 words, each containing 8 bits. The total capacity is 512 × 8 = 4096 bits, or 4 Kbit. Address and control signals select the word to be accessed. During a write, complementary data values are applied to BL and BLB and the access network is enabled to update the latch. During a read, the selected cell produces a bit-line response corresponding to the stored value. In hold mode, access is disabled and the cross-coupled storage structure retains its state while the supply remains present.

C. Static Noise Margin

SNM quantifies the maximum DC noise voltage tolerated at the storage nodes without changing the cell state. In the butterfly method, the voltage-transfer characteristics of the two cross-coupled inverter halves are plotted together; the side length of the largest square that fits between the curves is used as the SNM. RSNM characterizes stability while the bit lines and access devices participate in reading. HSNM characterizes retention with access disabled. WSNM characterizes the write condition and is interpreted in relation to the ability of applied data to overwrite the previous state.

The supplied bit-cell results are RSNM = 309.08 mV, HSNM = 430.81 mV and WSNM = 664.60 mV. These are transistor-level simulation values, not array-level timing, area or power metrics.

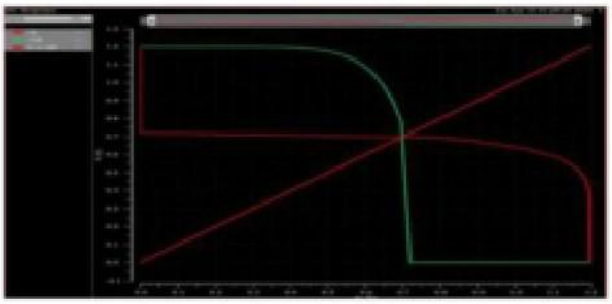


Fig. 2. DC butterfly-curve view used for static-noise-margin evaluation.

III. DESIGN AND IMPLEMENTATION METHODOLOGY

A. Transistor-Level Functional Simulation

The 12T schematic is simulated in Cadence Virtuoso. A pre-layout test bench applies data, address and control signals to exercise hold, read and write conditions. Waveforms are inspected to confirm that selected-word and control activity correspond to intended memory operations. The supplied SimVision capture shows address, clock, chip-select, read-enable, write-enable and data activity. It is qualitative evidence of functional stimulus and response; individual transition delays, read access time and energy per operation were not supplied.

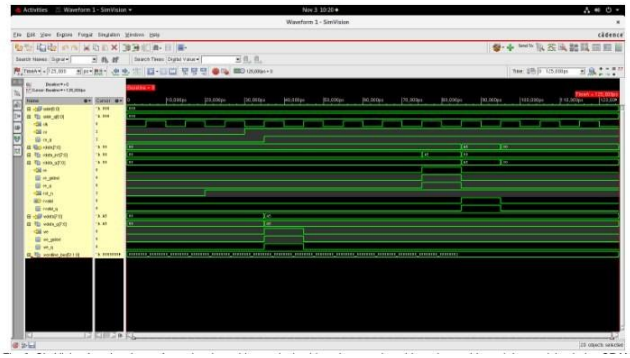


Fig. 3. SimVision waveform showing address, clock, chip-select, read-enable, write-enable and data activity.

B. RTL Synthesis and Physical-Design Entry

The memory RTL is synthesized using Cadence Genus to obtain a standard-cell implementation. The synthesized netlist and associated constraints are transferred to Cadence Innovus. Synthesis maps the RTL description into library cells; physical implementation assigns those cells to locations and creates the interconnect structures required by the design. The project evidence includes floorplan/core setup and subsequent implementation views.

C. Floorplanning and Placement

Floorplanning defines the core region in which standard-cell rows, power structures and signal interconnects are arranged. Placement assigns synthesized cells to legal locations while observing row structure and implementation constraints. Placement density and routability influence the resources available for power and signal routing. The supplied Innovus floorplan view documents the core region before power-grid and signal-routing details are displayed.

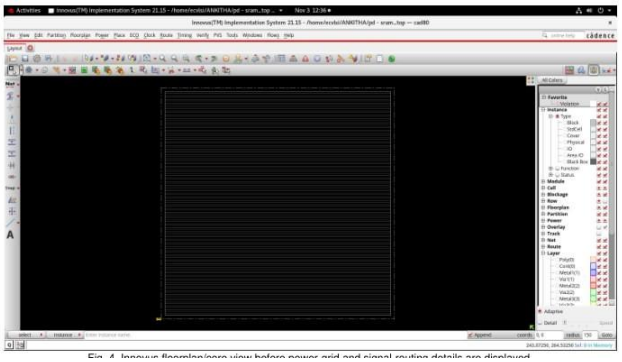


Fig. 4. Innovus floorplan/core view before power-grid and signal-routing details.

D. Power-Distribution Planning

After placement, the power-distribution network connects supply sources to the placed cells. Power rings and stripes provide paths across the core for supply and ground. The captured Innovus view shows an orthogonal stripe pattern extending across the core. This demonstrates the power-planning stage; it does not independently establish IR-drop or electromigration compliance.

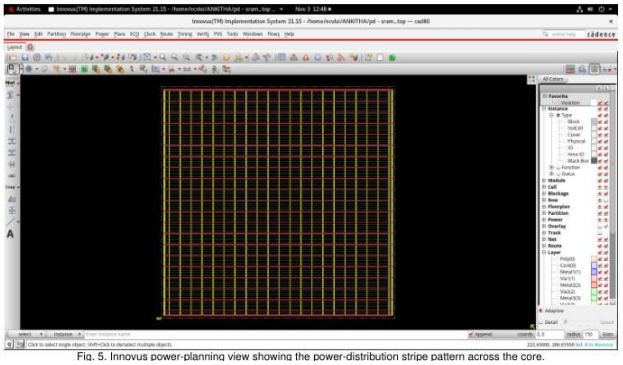


Fig. 5. Innovus power-planning view showing the power-distribution stripe pattern across the core.

Fig. 5. Innovus power-planning view showing the power-distribution stripe pattern.

E. Physical-Design Evidence and Interpretation

The floorplan, power-grid, routing and timing captures represent different checkpoints rather than interchangeable views of a completed design. The floorplan image shows the core region and the initial placement environment. The stripe view records the power-distribution planning stage. The routed view then shows signal interconnects over the core, while the timing capture reports a selected hold analysis. Keeping these checkpoints separate is important when documenting implementation progress: the presence of a later-stage screenshot does not automatically prove that every earlier or later verification task has passed.

The reported design density of 85.981% is associated with the captured timing summary. Density is a measure of how much of the available placement region is occupied by cells under the tool's reporting convention; it is not itself a measure of routing completion or timing quality. A high utilization can reduce available whitespace for routing and optimization, while a lower utilization can increase core area. The supplied evidence does not include congestion maps or utilization targets, so no claim about optimal density is made.

The hold summary reports WNS of 0.032 ns and TNS of 0.000 ns across 38 paths, with zero violating paths in the displayed report. WNS describes the worst slack among the paths included in the selected analysis, whereas TNS aggregates negative slack for violating paths. These quantities are mode- and constraint-dependent. The reported zero violations therefore applies only to the displayed hold analysis and does not establish setup closure, all-corner closure or sign-off timing.

Power and area figures also require their analysis context. Internal power is associated with switching activity internal to cells, switching power is associated with charging and discharging interconnect capacitances, and leakage power is the static component reported by the analysis. The project summary lists 95.42 mW internal power, 5.239 mW switching power and 1.637 µW leakage power. Because the activity file, voltage, frequency, process corner and power-analysis stage are not included, these values should be treated as the reported project outputs rather than universal characteristics of the 4-Kbit architecture.

IV. ROUTING AND IMPLEMENTATION RESULTS

A. Signal Routing

Signal routing connects placed standard-cell pins according to the synthesized netlist. It is distinct from power-stripe insertion: signal routing implements logical net connectivity, whereas the power network distributes supply and ground. The supplied Innovus capture shows the routed interconnect network across the core. This is visual evidence of routing activity, not a substitute for detailed route reports, parasitic extraction or sign-off checks.

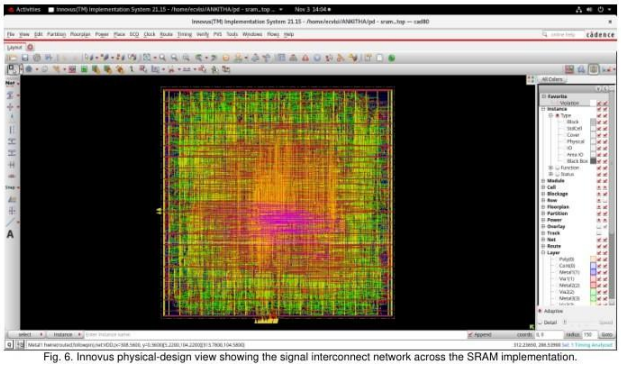


Fig. 6. Innovus physical-design view showing the signal interconnect network across the SRAM implementation.

Fig. 6. Innovus physical-design view showing the signal interconnect network.

B. Timing Summary

The available terminal capture shows an Innovus timeDesign hold-mode summary. It reports WNS = 0.032 ns, TNS = 0.000 ns and zero violating paths, with 38 paths included in the displayed summary. The same capture reports design density of 85.981%. These figures describe the specific hold-mode analysis shown in the project material. They should not be interpreted as complete post-route setup-and-hold sign-off because the corresponding full reports and analysis conditions were not provided.

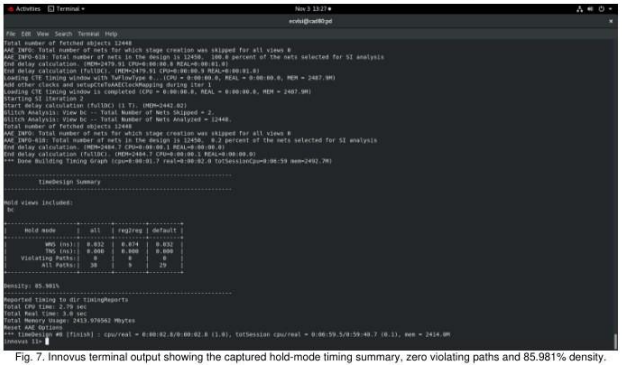


Fig. 7. Innovus terminal output showing the captured hold-mode timing summary, zero violating paths and 85.981% density.

Fig. 7. Captured Innovus hold-mode timing summary and reported design density.

C. Implementation Metrics

The separate implementation summary reports 906 standard cells and a total area of 413867.378; the area unit is not identified. Total power is 100.66 mW, comprising 95.42 mW internal power and 5.239 mW switching power. Leakage power is 1.637 µW. The project material also lists a 4-ns clock period and a 2.512-ns data path. Since the exact analysis view, corner, activity assumptions and area unit are not included, these values are reproduced without additional interpretation.

D. Limits of the Available Implementation Metrics

The reported cell count and area should be read with the same care as the power and timing values. The cell count of 906 describes the mapped standard-cell implementation captured by the project summary; it does not identify the number of SRAM bit cells, the number of transistors in the custom 12T cell, or the capacity of the array by itself. The area value 413867.378 is reproduced without a unit because the supplied output does not state whether it is expressed in square micrometres, database units or another measure. Assigning a unit without the implementation report would create an unsupported result.

The project summary also lists a 4-ns clock period and a 2.512-ns data path. These values are retained as reported, but the available material does not identify the exact path endpoints, timing path type, clock uncertainty, input/output delays or operating corner. A data-path value alone is not sufficient to calculate maximum operating frequency or to establish setup timing closure. A complete timing appendix should include the full path report, clock definition, constraints and process-voltage-temperature.

For a complete physical-design evaluation, the next verification steps should include post-route parasitic extraction, setup and hold analysis across required corners, DRC, LVS, antenna checks, IR-drop analysis and electromigration assessment. These reports would confirm whether the routed geometry conforms to the selected design rules and whether power delivery remains reliable under the intended operating conditions. They are recommended follow-up checks, not results claimed by the present project evidence.

E. Interpretation of the Reported Results

The reported static-noise margins describe the tolerance of the simulated 12T bit cell under three different operating conditions. The read margin is 309.08 mV, while the hold margin is 430.81 mV. This difference is consistent with the fact that read access activates the access network and couples the storage nodes to the bit lines, whereas the hold condition isolates the latch from the access path. The write margin is reported as 664.60 mV for the supplied write test setup. These values should be treated as results of the stated transistor-level simulation; they do not, by themselves, establish the behavior of every cell in a fabricated memory array.

F. Relationship Between Cell and Array Evaluation

Two levels of evaluation are involved in this project. At the bit-cell level, the transistor schematic and DC butterfly curves are used to study storage stability. The 12-transistor topology includes the cross-coupled latch, access devices and additional feedback devices. At the array implementation level, the 512-word by 8-bit organization is handled through an RTL synthesis and physical-design flow. The standard-cell count and implementation reports therefore describe the mapped digital design and must not be interpreted as the transistor count or physical area of one custom SRAM bit cell.

G. Interpretation of Physical-Design Checkpoints

Each implementation view documents a particular stage of the flow. The floorplan view establishes the core region and placement environment. The power-planning view shows the distribution of supply and ground structures across that region. The routed view documents signal interconnections between placed cells. The timing capture is a report for a selected hold-mode analysis. These views provide complementary evidence, but one view cannot replace the reports associated with another stage. In particular, a visible routed network does not provide the extracted parasitics needed for a complete post-route timing assessment.

H. Scope of Verification

The displayed hold summary includes 38 paths and reports WNS of 0.032 ns, TNS of 0.000 ns and zero violating paths. This is a positive result for the analysis represented by that capture, but it is limited to the selected mode and constraints. A complete assessment would also require the corresponding setup analysis, relevant process-voltage-temperature corners and the final timing constraints. The available material does not provide those full reports, so no claim of all-corner timing closure is made.

I. Reporting and Reproducibility

For results to be reproduced or compared fairly, the report should identify the device models, supply voltage, temperature, transistor dimensions and SNM extraction procedure used for the bit-cell simulations. The array-level results should identify the standard-cell library, clock definition, input and output constraints, timing corner, extraction stage and switching-activity assumptions used for power estimation. The area unit should also be stated in the implementation report. Since some of these details are not present in the supplied project evidence, the numerical values in this paper are retained as reported and are not assigned additional assumptions.

V. GDS LAYOUT EVIDENCE AND DISCUSSION

The supplied Cadence Design Review screenshot displays the generated file sram_top.gds and its integrated layout geometry. This establishes that a GDS view was generated and opened for inspection. A GDS viewer image alone does not establish that the layout has passed design-rule checking (DRC), layout-versus-schematic (LVS), antenna, IR-drop or electromigration verification. Such claims require the corresponding final verification reports.

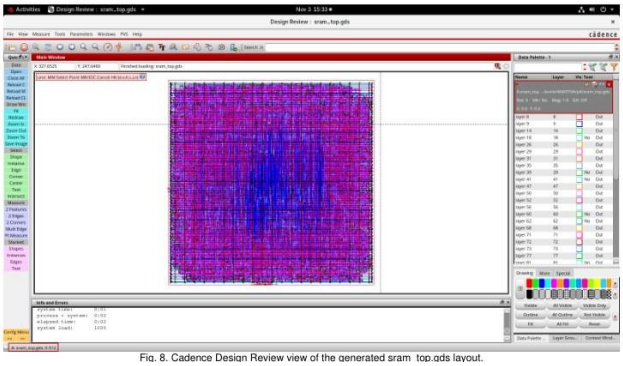


Fig. 8. Cadence Design Review view of the generated sram_top.gds layout.

Fig. 8. Cadence Design Review view of the generated sram_top.gds layout.

The results fall into two distinct domains. Static-noise margins are properties of the simulated transistor-level bit cell under its test conditions. Cell count, area, power, density and timing are reported for the synthesized and physically implemented memory design. Combining these categories into one performance claim would obscure the difference between circuit-level characterization and array-level implementation.

The three operating modes exercise different electrical conditions and should be considered separately. In hold mode, word-line access devices are disabled and the cross-coupled storage network preserves the stored state. During read, the access path is active and internal storage nodes may experience disturbance as bit lines develop their response. The RSNM of 309.08 mV is lower than the hold margin of 430.81 mV, illustrating why read stability must be assessed independently. During write, the access network must allow externally driven data to change the latch state. The reported WSNM is 664.60 mV under the project's write test setup. Direct comparison with another cell requires matching supply voltage, device models, sizing, temperature and measurement procedure.

A bit-cell simulation and a memory-array implementation answer different questions. The transistor-level test bench examines an individual storage element and supports extraction of its butterfly curves. The array-level RTL flow evaluates the digital memory description after synthesis and physical implementation. At this level, mapped standard-cell count and reported area depend on RTL structure, synthesis mapping, library cells and implementation constraints. Power estimates also depend on analysis setup and switching assumptions. The 12T transistor count must not be equated with the reported 906 standard cells: the latter is a mapped implementation count for the synthesized design, not the transistor count of one memory bit cell.

The order of physical-design operations matters because each stage creates inputs and constraints for the next. Floorplanning establishes the core boundary and placement region. Placement determines cell locations and influences wire lengths and congestion. Power planning adds supply structures that consume routing resources, so power stripes and signal routes must coexist within the available interconnect layers. Signal routing completes net connections represented by the netlist. Timing analysis evaluates whether implemented paths satisfy the constraints for the selected mode.

The available timing capture reports no violating paths in the 38-path hold summary. However, complete setup timing, post-route parasitic, DRC, LVS, IR-drop and electromigration reports were not supplied. The unspecified area unit and missing power-analysis conditions also limit direct comparison with other implementations. A reproducible report should identify the process/device-model version, supply voltage, temperature, transistor dimensions, SNM extraction procedure, standard-cell library, clock and I/O constraints, timing corner, extraction stage and power activity assumptions. These details are not fully present in the supplied project material, so the paper records the available evidence without filling those gaps by assumption.

VI. CONCLUSION

A Noble Schmitt Trigger-based 12T SRAM bit cell and a 512 × 8 (4-Kbit) memory organization have been studied in a 90-nm technology environment. The supplied project material documents the transistor-level schematic, butterfly-curve stability evaluation and functional read/write waveform, followed by RTL synthesis and the physical-design sequence of floorplan/core setup, placement, power-stripe planning, signal routing, timing review and GDS generation. The bit-cell simulation reports RSNM, HSNM and WSNM values of 309.08 mV, 430.81 mV and 664.60 mV. The implementation summary reports 906 standard cells, area 413867.378 (unit unspecified), total power 100.66 mW and leakage power 1.637 µW. The captured hold-mode summary shows zero violating paths. Complete post-route setup/hold reports, extracted parasitics, DRC/LVS results and power-integrity verification should be included before final sign-off claims are made.

REFERENCES

[1] E. Seevinck, F. J. List and J. Lohstroh, “Static-noise margin analysis of MOS SRAM cells,” *IEEE Journal of Solid-State Circuits*, vol. SC-22, no. 5, pp. 748–754, Oct. 1987.

[2] A. J. Bhavnagarwala, X. Tang and J. D. Meindl, “The impact of intrinsic device fluctuations on CMOS SRAM cell stability,” *IEEE Journal of Solid-State Circuits*, vol. 36, no. 4, pp. 658–665, Apr. 2001, doi: 10.1109/4.913744.

[3] N. H. E. Weste and D. Money Harris, *CMOS VLSI Design: A Circuits and Systems Perspective*, 4th ed. Addison-Wesley, 2011.

[4] J. M. Rabaey, A. Chandrakasan and B. Nikoli■, *Digital Integrated Circuits: A Design Perspective*, 2nd ed. Prentice Hall, 2003.

[5] Cadence Design Systems, “Virtuoso, Genus and Innovus implementation screenshots and reports supplied with the project materials,” 2026.

[6] S. R. Mansore, R. S. Gamad and D. K. Mishra, “Schmitt-Trigger Based 12T SRAM Cell with Enhanced Stability,” *Journal of Semiconductor Devices and Circuits*, vol. 5, no. 1, pp. 5–10, 2018, doi: 10.37591/josdc.v5i1.549.

[7] “A low power subthreshold Schmitt Trigger based 12T SRAM bit cell with process-variation-tolerant write-ability,” *Microelectronics Journal*, vol. 97, Art. 104703, Mar. 2020, doi: 10.1016/j.mejo.2020.104703.

[8] “Read Stability and Write-Ability Analysis of SRAM Cells for Nanometer Technologies,” *IEEE Journal of Solid-State Circuits*, vol. 41, no. 11, pp. 2577–2588, Nov. 2006, doi: 10.1109/JSSC.2006.883344.

[9] “The Stability Performance Analysis of SRAM Cell Topologies in 90nm and 130nm CMOS technology,” in *Proc. 2021 International Conference on Emerging Smart Computing and Informatics (ESCI)*, Pune, India, Mar. 2021, doi: 10.1109/ESCI50559.2021.9396973.

[10] “Design and Analysis of Schmitt Trigger Based 10T SRAM in 32 nm Technology,” in *Proc. 2017 International Conference on Innovations in Information, Embedded and Communication Systems*, 2017, doi: 10.1109/INIS.2017.56.